AYUSH JHA

Bengaluru, India | P: +91 8296526863 | ashujha301@gmail.com | Github | LinkedIn | Portfolio

WORK EXPERIENCE

SORT-TREE. (Remote) London, UK
AI/ML Engineer (Contract) Nov 25 – Present

- Working as a Machine Learning Engineer on a performance marketing platform, building ML-driven systems for product analytics and campaign performance optimization.
- Developed Python (Fast API) backend services integrating ML outputs into production dashboards; ingested and processed data from Meta/Facebook & Instagram Graph APIs for product analytics and competitor intelligence.
- Analytics dashboard using Next.js (frontend) and Python (backend), enabling real-time monitoring of campaign metrics and data-driven decision-making.

SARANYU TECHNOLOGIES. Bengaluru, Karnataka
Software Engineer Sep 24 – Oct 25

- Working on client projects, contributing to frontend development using Next.js, backend API creation using Node.js (Fastify) and PHP to support scalable web applications.
- Working with CI/CD GitHub Actions for automated deployment to improve development workflow. Collaborating with the team using Git for version control and efficient project management.
- Actively participating in agile methodologies, including sprint planning and performing code reviews through GitHub pull requests.

APPKNOX. Bengaluru, Karnataka
Software Engineering Intern May 24 – Aug 24

- Added SARIF report generation to AppKnox CLI, enabling GHAS integration. Worked on GitHub Actions for CI/CD automation, ensuring seamless software delivery.
- Developed and integrated a Jenkins CI/CD plugin, automating security scanning for 200+ enterprise clients.
- Engineered a Python-based web scraper to systematically collect and store metadata for 15,000+ Play Store apps, enhancing data processing efficiency by 35%.

PROJECTS

ML MODELS.

ML Models from Scratch: - Production-Style ML Systems (Python, Fast API, Kafka, PostgreSQL)

- Developed three real-world ML systems: **Marketing & Housing Price Prediction (Linear Regression)**, **Real-Time Fraud Detection (Logistic Regression)**, and **Loan Approval Decision System (Decision Tree)**.
- Designed end-to-end ML pipelines including PostgreSQL data ingestion, feature engineering, custom gradient descent training, model persistence, and structured inference workflows without using pre-built ML libraries.
- Built production-style architecture using Fast API (concurrent APIs), Kafka (streaming), WebSocket (real-time output), Prometheus & Grafana (monitoring), and implemented retraining, logging, and model versioning with rollback support.

ZYNK – AI POWERED VOICE AND GESTURE CHROME EXTENTION.

Intelligent Browser Automation System (Computer Vision + Voice AI, Chrome Web Store)

- Built and published a full-stack AI Chrome extension enabling voice- and gesture-controlled browser automation; integrated faster-whisper for on-device speech recognition with Voice Activity Detection (VAD) via Web Audio API for sub-100ms command execution latency.
- Deployed a production-grade backend on GCP (e2-micro VM, pm2 process management) with FastAPI, Supabase (PostgreSQL + pgBouncer connection pooling, SSL), in-memory Bearer token auth with silent refresh, rate limiting to control OpenAI API costs, and CORS-configured Vercel-deployed Next.js frontend.
- Implemented a RAG pipeline using LangChain, ChromaDB, and OpenAI Embeddings over the Zynk codebase as a corpus, enabling semantic understanding of voice commands, LLM-assisted automation logic, and LangSmith observability.

EDUCATION

REVA UNIVERSITY Bengaluru, Karnataka
Bachelor of Engineering Jul 2019 - Jul 2023

Major ECE; Minor Computer Science.
CGPA: 8.3 /10.

SKILLS

Programming Languages: Python, C++, Typescript
Machine Learning: TensorFlow, Pytorch, Scikit-learn, Data Preprocessing, Model Deployment, MlOps, ML Pipelines
Backend: Node.js, Python (Django, Fast API), NoSQL, MySQL, Redis, Vector DBs
Familiar: Git, Github, Cloud (AWS, GCP), Kafka, Jenkins, Docker, Kubernetes, MlFlow, Hugging face, LLM, RAG

ADDITIONAL

Certifications & Training: Deep Learning Specialization(Deeplearning.AI), Machine Learning Specialization (Deeplearning.AI) , Airtribe Backend Launchpad ,Tensor Flow Applications